

4.2 Ejercicios VECTORES

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Fundamentos de Programación

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1. Ejercicio 1

1.1. Primeros 2 pasos

1) Definición

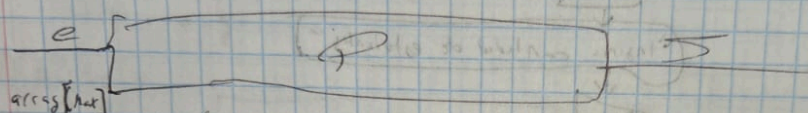
Count $\leftarrow 0$

~~array~~ $a \% 2 = 0$ $\hookrightarrow$ True

$a_i > \text{odd}$
 $\text{odd} \leftarrow a_i$

$a_i \% 2 = 0 \ \&\& \ \text{max} < a_i$
 $\text{max} \leftarrow a_i$
 $\text{pos} \leftarrow i$

2) Analisis



for ($i \leftarrow 0$; $i < \text{MAX}$; $i++$)
 $\text{sum} \leftarrow \text{array}[i]$

Mostrar: Suma posiciones pares

for ($i \leftarrow 1$; $i < \text{MAX}$; $i++$)
if ($\text{array}[i] > \text{max_odd}$)
 $\text{max_odd} \leftarrow \text{array}[i]$

Mostrar: Mayor posición impar

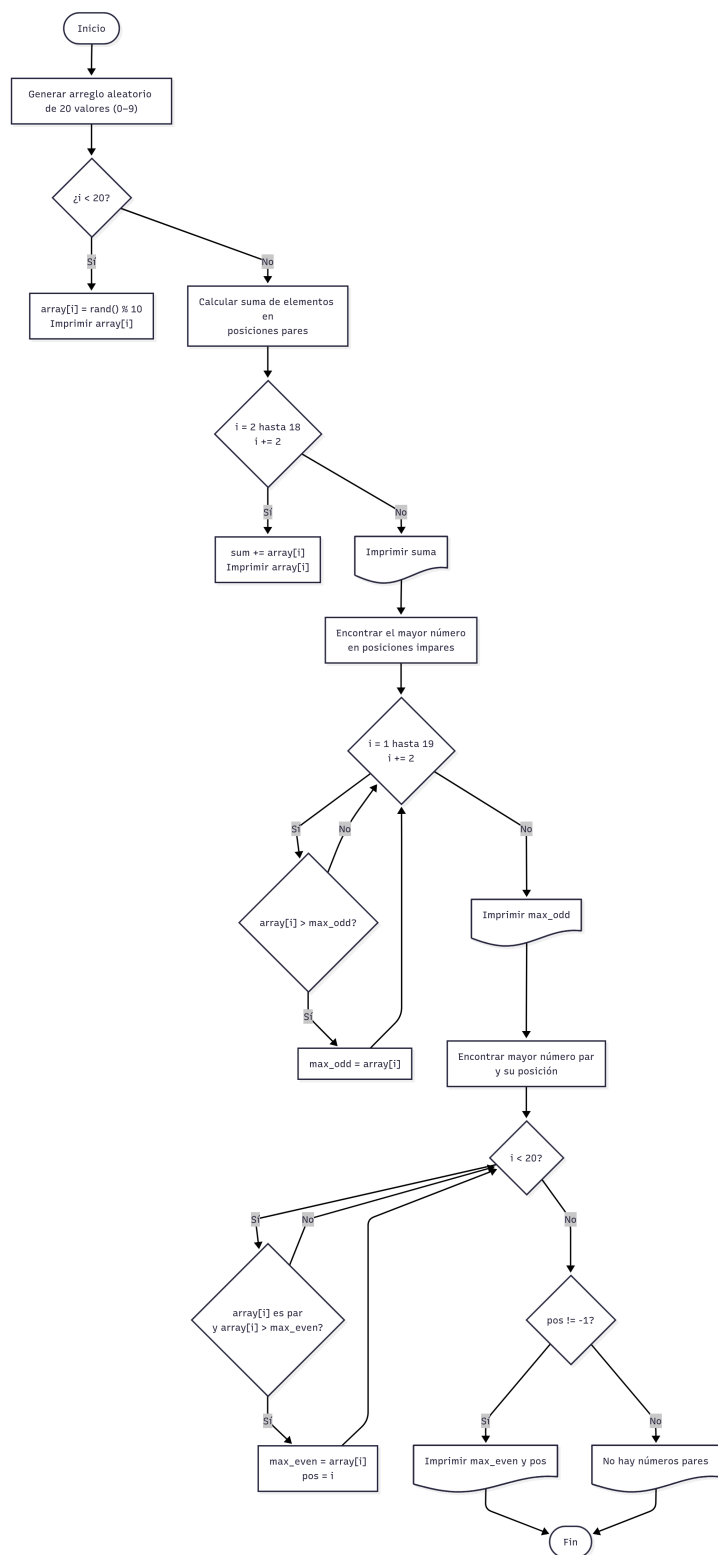
for ($i \leftarrow 0$; $i < \text{MAX}$; $i++$)
if ($\text{array}[i] \% 2 = 0 \ \&\& \ \text{array}[i] > \text{max}$)
 $\text{max} \leftarrow \text{array}[i]$
 $\text{pos} \leftarrow i$

Mostrar: Posición mayor número par

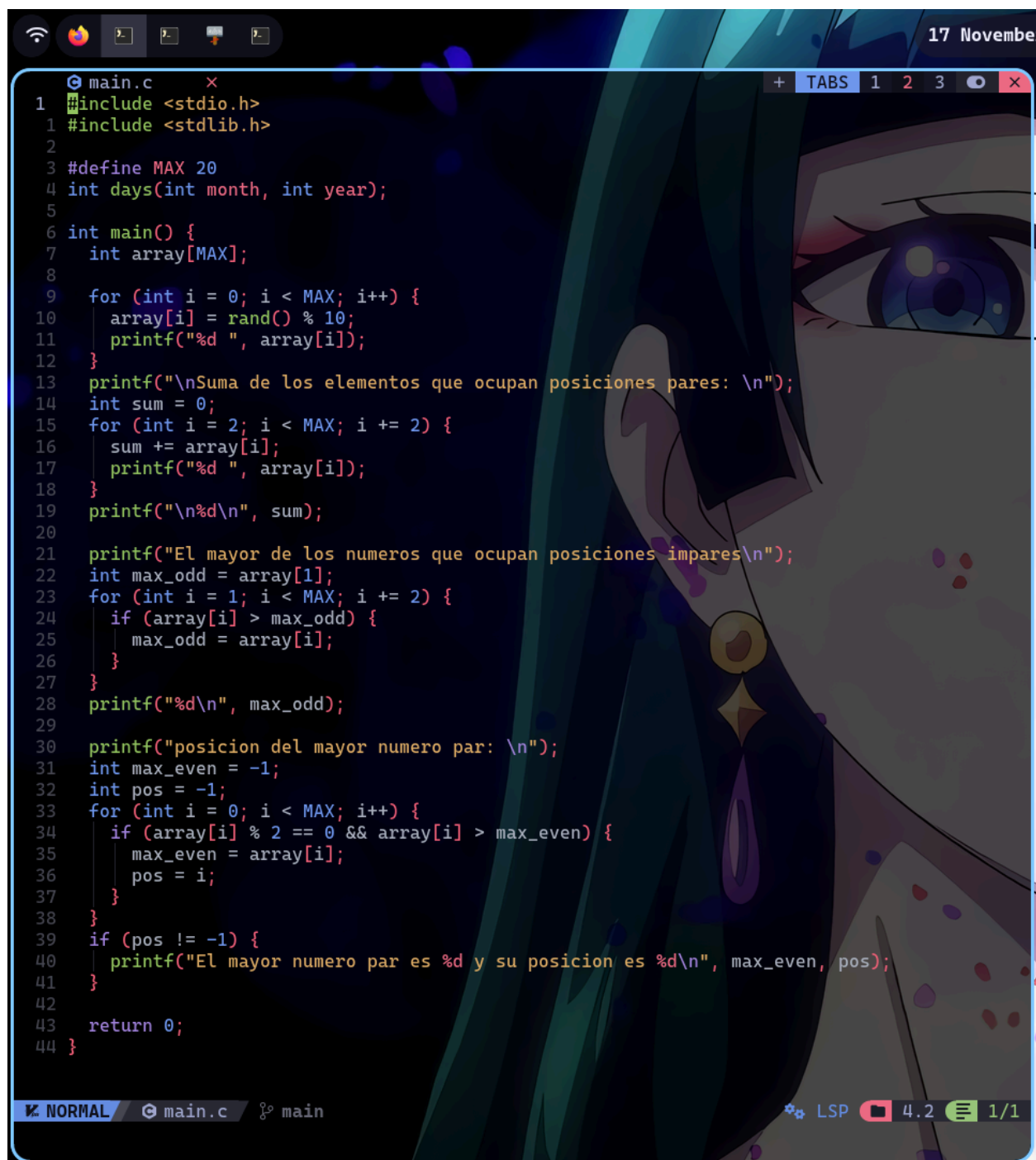
3) Algoritmo

Variable	Tipo	Comentario
array[]	Real	Array principal
sum	Real	Sumatoria posiciones pares
max_odd	Real	Mayor posición impar
max_even	Real	Mayor número par
pos	Real	Posición del mayor número par

1.2. Diagrama de flujo

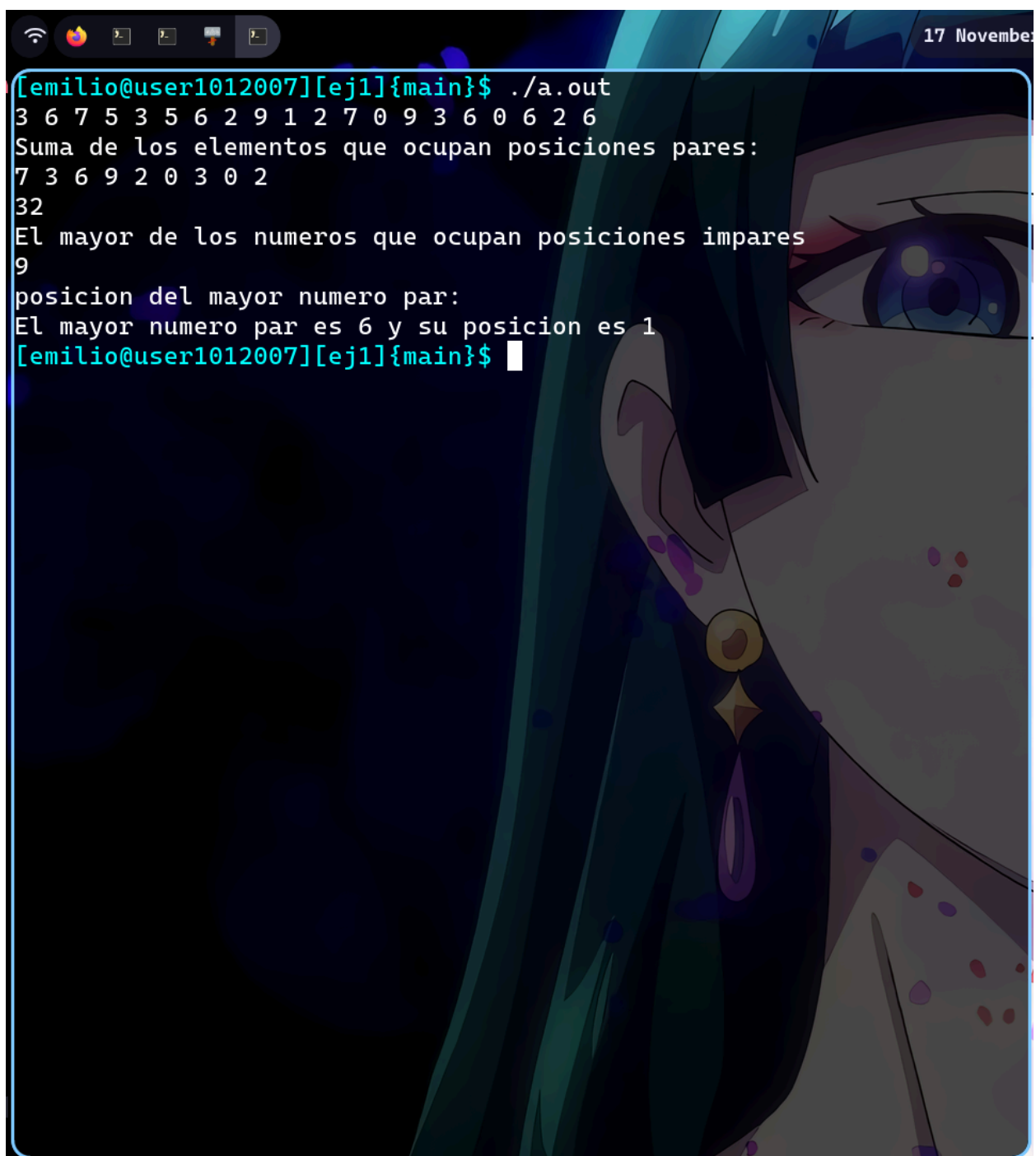


1.3. Codigo



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 20
5 int days(int month, int year);
6
7 int main() {
8     int array[MAX];
9
10    for (int i = 0; i < MAX; i++) {
11        array[i] = rand() % 10;
12        printf("%d ", array[i]);
13    }
14    printf("\nSuma de los elementos que ocupan posiciones pares: \n");
15    int sum = 0;
16    for (int i = 2; i < MAX; i += 2) {
17        sum += array[i];
18        printf("%d ", array[i]);
19    }
20    printf("\n%d\n", sum);
21
22    printf("El mayor de los numeros que ocupan posiciones impares\n");
23    int max_odd = array[1];
24    for (int i = 1; i < MAX; i += 2) {
25        if (array[i] > max_odd) {
26            max_odd = array[i];
27        }
28    }
29    printf("%d\n", max_odd);
30
31    printf("posicion del mayor numero par: \n");
32    int max_even = -1;
33    int pos = -1;
34    for (int i = 0; i < MAX; i++) {
35        if (array[i] % 2 == 0 && array[i] > max_even) {
36            max_even = array[i];
37            pos = i;
38        }
39    }
40    if (pos != -1) {
41        printf("El mayor numero par es %d y su posicion es %d\n", max_even, pos);
42    }
43    return 0;
44 }
```

1.4. Prueba de escritorio

A terminal window with a dark background and an anime-style illustration of a character's face on the right. The terminal shows the execution of a program that processes an array of numbers. The output includes the array itself, the sum of elements at even indices, the maximum value at an odd index, and its position.

```
[emilio@user1012007][ej1]{main}$ ./a.out
3 6 7 5 3 5 6 2 9 1 2 7 0 9 3 6 0 6 2 6
Suma de los elementos que ocupan posiciones pares:
7 3 6 9 2 0 3 0 2
32
El mayor de los numeros que ocupan posiciones impares
9
posicion del mayor numero par:
El mayor numero par es 6 y su posicion es 1
[emilio@user1012007][ej1]{main}$
```

2. Ejercicio 1

2.1. Primeros 2 pasos

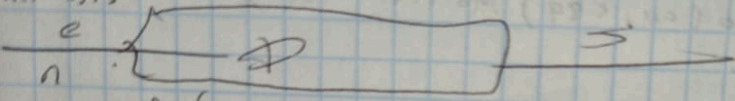
1) Definición

$$\text{suma} \leftarrow a_i \% 2 = 0 ? a_i : 0$$

$$\text{suma} \leftarrow a_i \% 2 \neq 0 ? a_i : 0$$

$$\text{suma} \leftarrow \sum_{i=0}^n a_i$$

2) Análisis



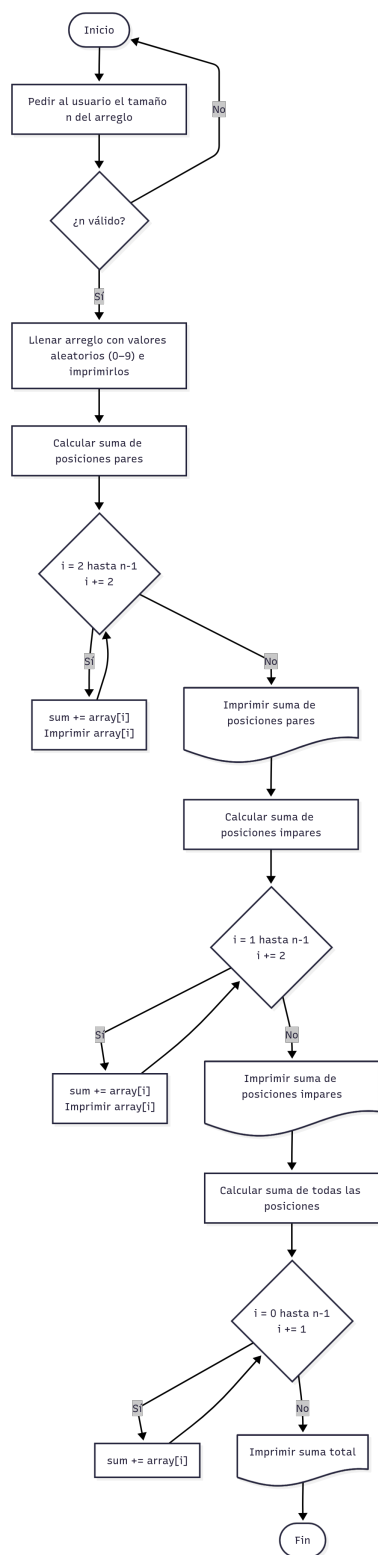
```

for (i = 2; i < n; i += 2)    mostrar todos los
    suma += array[i]         = incluye
for (i = 1; i < n; i++)
    suma += array[i]
for (i = 1; i < n; i++)
    suma += array[i]
  
```

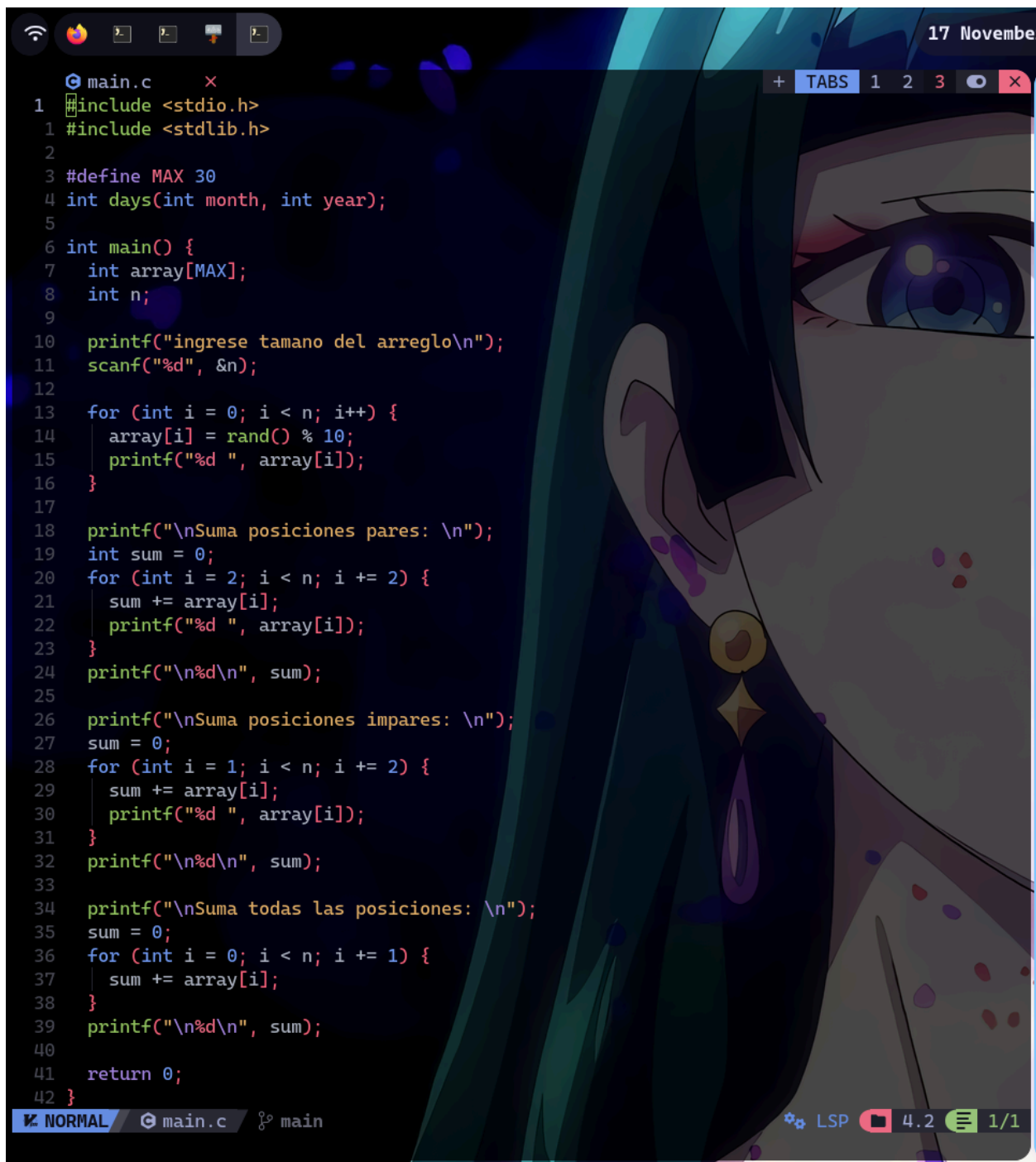
3) Algoritmo

Variable	Tipo	Construcción
n	Real	Tamaño del arreglo
suma 1	Real	suma de todos los pares
suma 2	Real	$\sum$ posiciones impares
suma 3	Real	$\sum$ todos los elementos
array	Real	arreglo principal

2.2. Diagrama de flujo



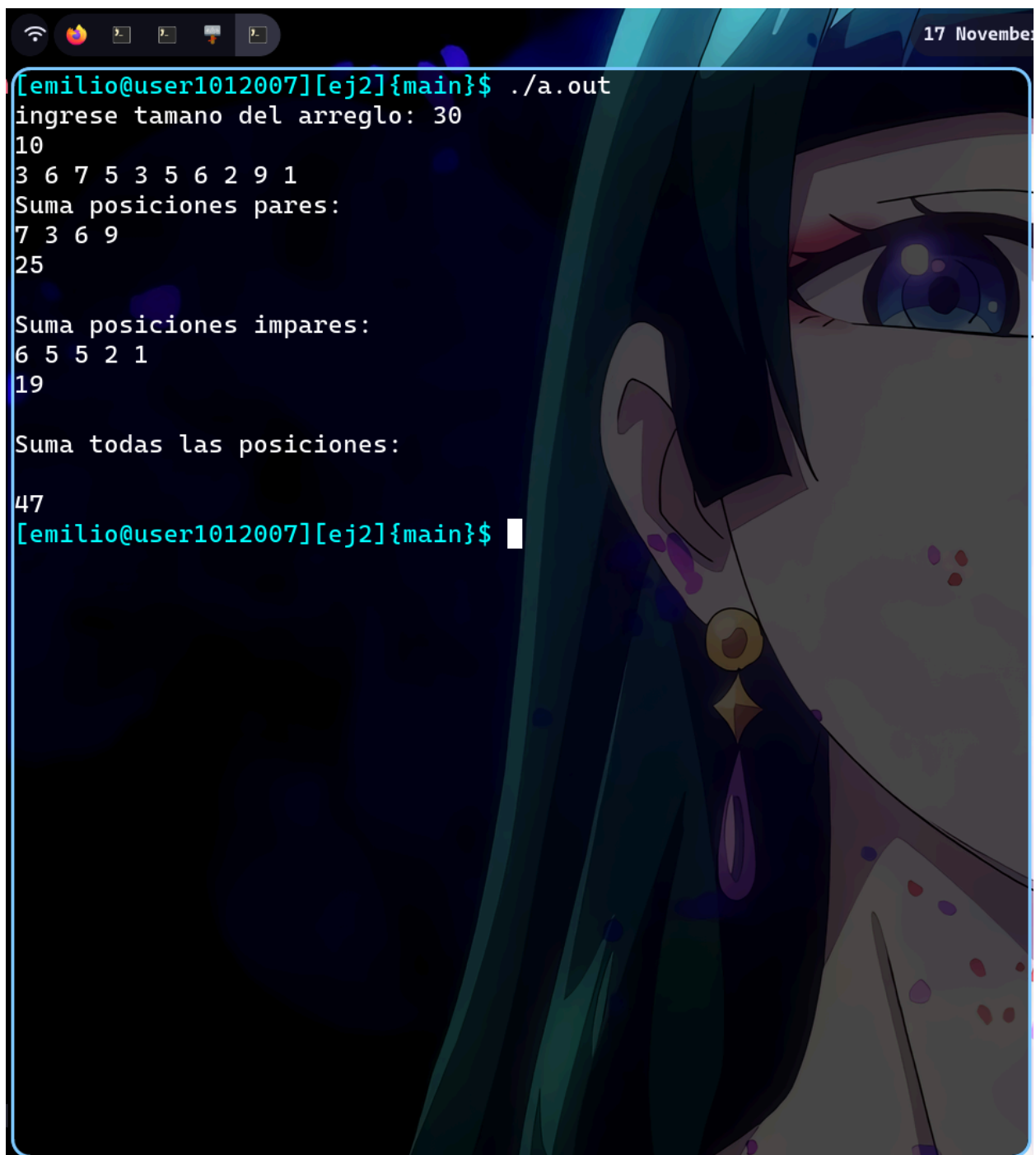
2.3. Código



```
1 #include <stdio.h>
1 #include <stdlib.h>
2
3 #define MAX 30
4 int days(int month, int year);
5
6 int main() {
7     int array[MAX];
8     int n;
9
10    printf("ingrese tamaño del arreglo\n");
11    scanf("%d", &n);
12
13    for (int i = 0; i < n; i++) {
14        array[i] = rand() % 10;
15        printf("%d ", array[i]);
16    }
17
18    printf("\nSuma posiciones pares: \n");
19    int sum = 0;
20    for (int i = 2; i < n; i += 2) {
21        sum += array[i];
22        printf("%d ", array[i]);
23    }
24    printf("\n%d\n", sum);
25
26    printf("\nSuma posiciones impares: \n");
27    sum = 0;
28    for (int i = 1; i < n; i += 2) {
29        sum += array[i];
30        printf("%d ", array[i]);
31    }
32    printf("\n%d\n", sum);
33
34    printf("\nSuma todas las posiciones: \n");
35    sum = 0;
36    for (int i = 0; i < n; i += 1) {
37        sum += array[i];
38    }
39    printf("\n%d\n", sum);
40
41    return 0;
42 }
```

The screenshot shows a code editor with a dark theme. The code is a C program named 'main.c'. It includes standard headers, defines a constant MAX as 30, and declares a function 'days'. The 'main' function prompts the user for the size of an array 'n', generates random numbers for the array, and then calculates and prints the sum of even-indexed elements, odd-indexed elements, and the sum of all elements. The editor interface includes a top bar with system icons and a date '17 November', a tab bar with 'main.c' selected, and a bottom status bar showing 'NORMAL', 'main.c', 'main', 'LSP', '4.2', and '1/1'.

2.4. Prueba de escritorio



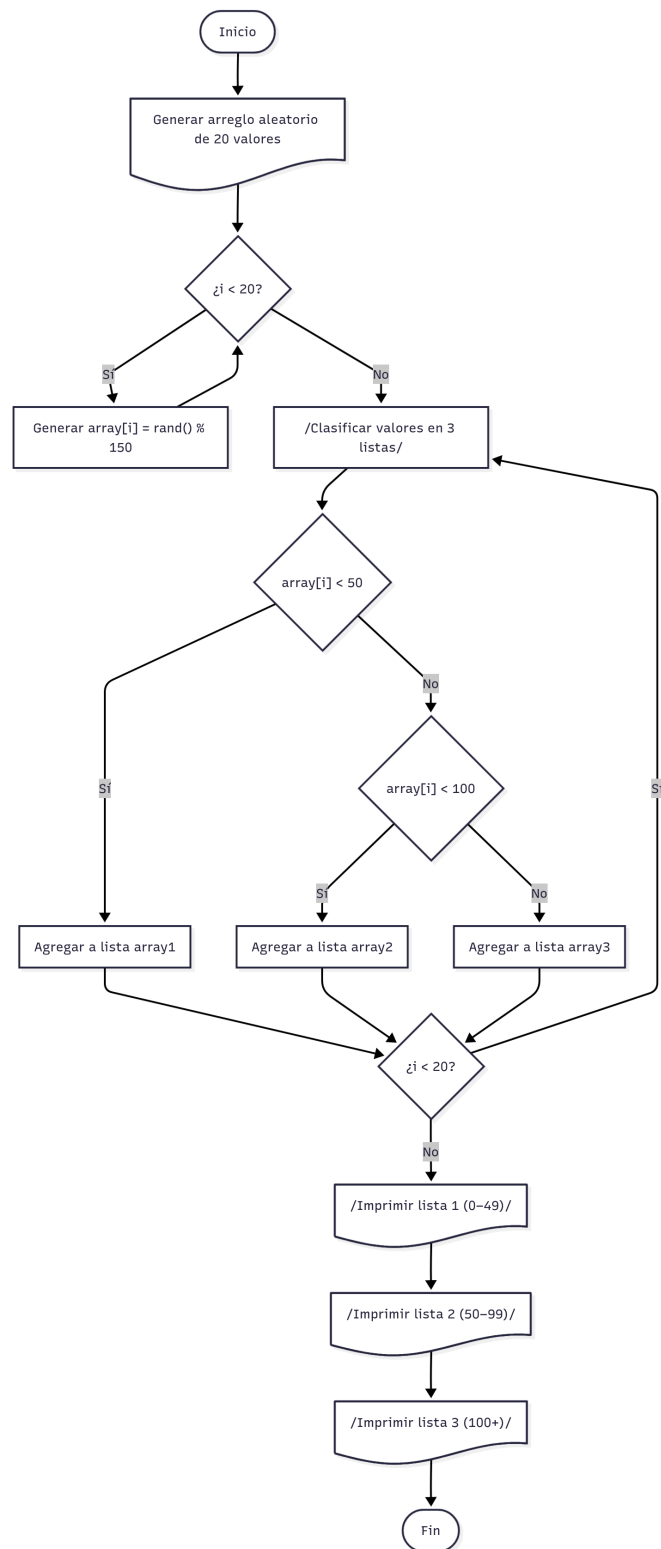
A terminal window with a dark background and an anime-style illustration of a character's face on the right. The terminal shows the execution of a program named `a.out`. The user enters the array size 30, and the program outputs the array `3 6 7 5 3 5 6 2 9 1`. It then calculates the sum of even-indexed elements (7, 3, 6, 9) resulting in 25, and the sum of odd-indexed elements (6, 5, 5, 2, 1) resulting in 19. Finally, it outputs the total sum of all elements as 47.

```
[emilio@user1012007][ej2]{main}$ ./a.out
ingrese tamano del arreglo: 30
10
3 6 7 5 3 5 6 2 9 1
Suma posiciones pares:
7 3 6 9
25

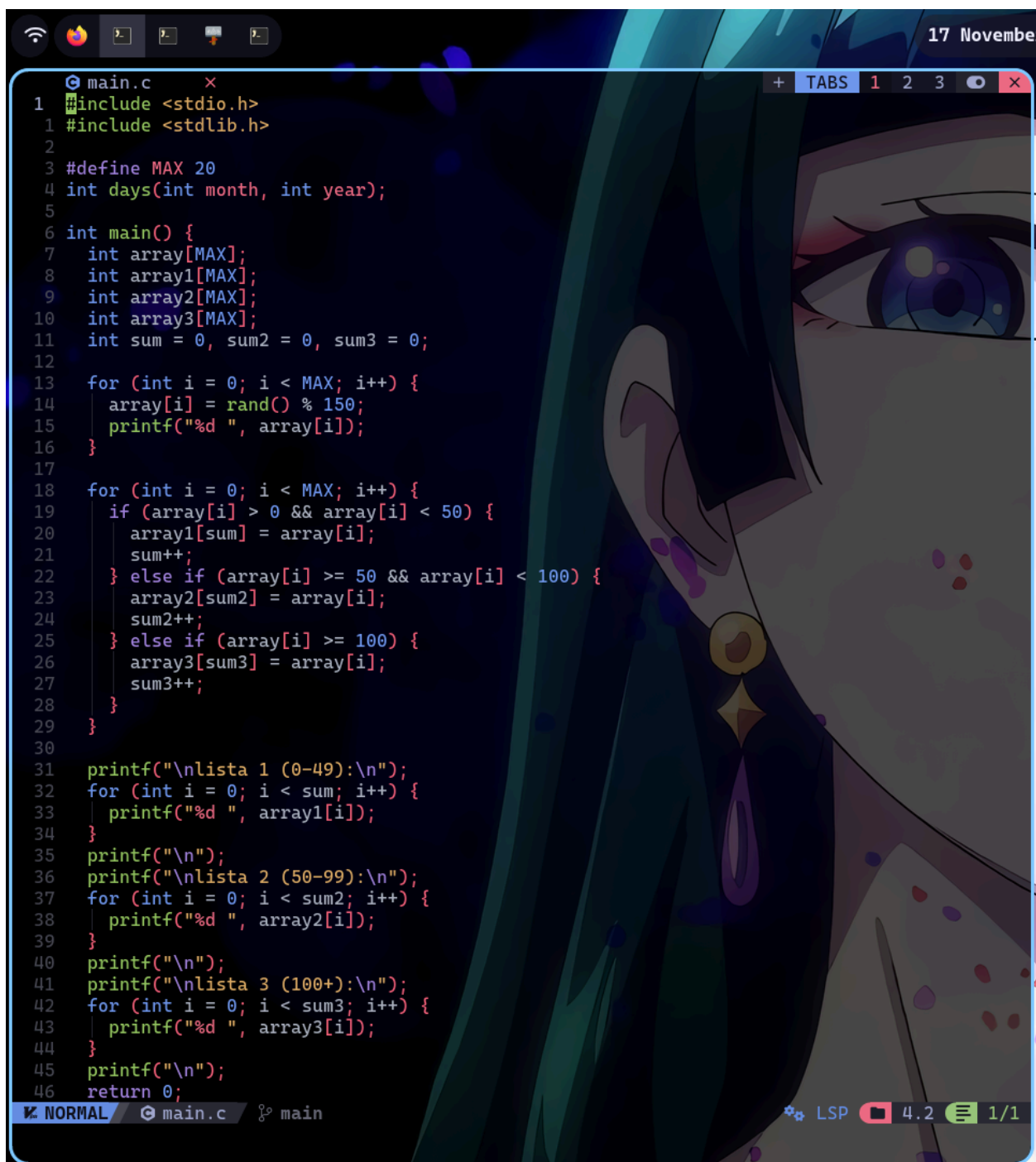
Suma posiciones impares:
6 5 5 2 1
19

Suma todas las posiciones:
47
[emilio@user1012007][ej2]{main}$
```


3.2. Diagrama de flujo

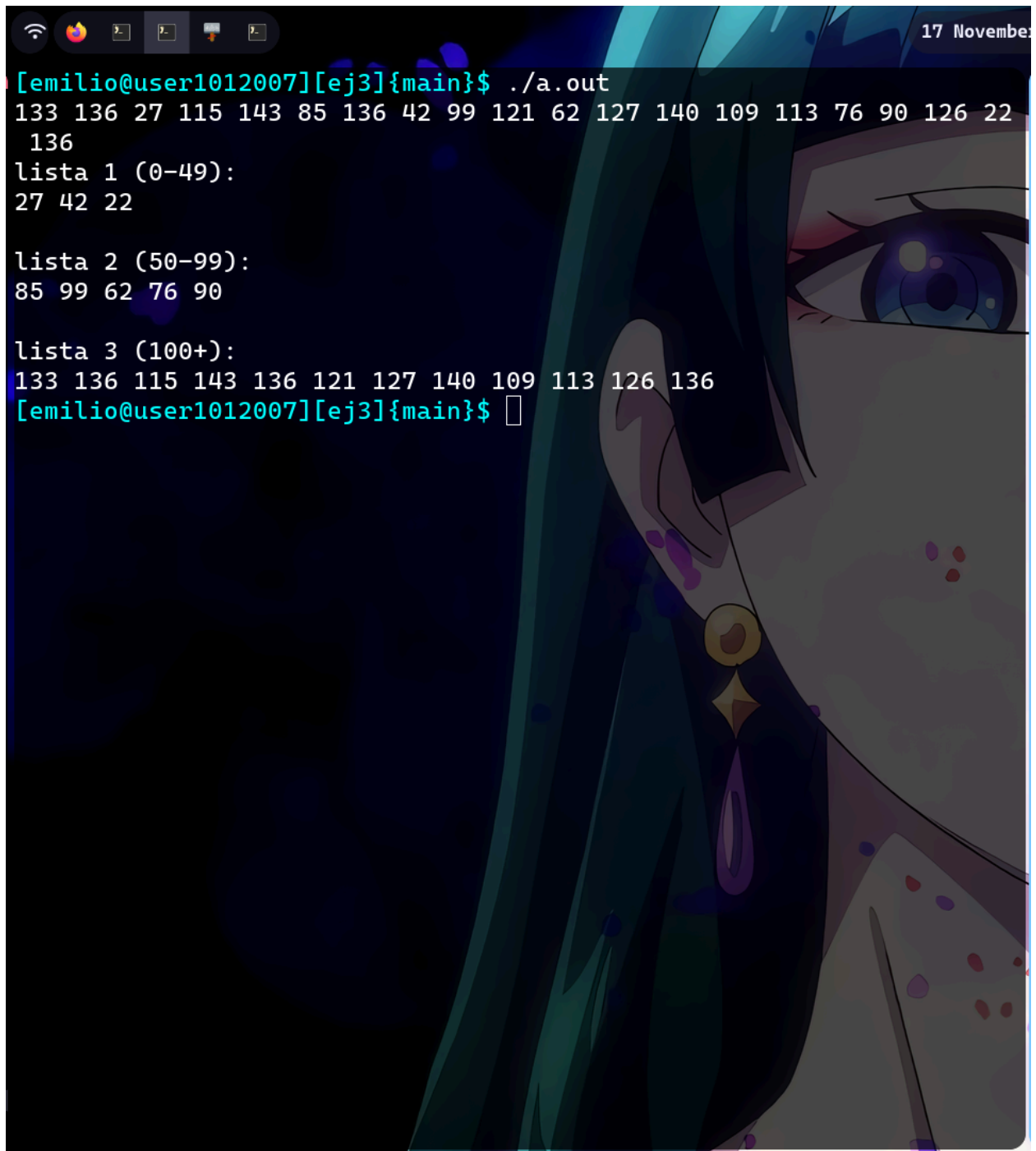


3.3. Código

A screenshot of a code editor window with a dark theme. The editor shows a C program named 'main.c'. The code generates an array of 20 random numbers and then classifies them into three categories based on their value: 0-49, 50-99, and 100+. The background of the editor features a stylized anime-style character with blue eyes and dark hair. The status bar at the bottom indicates 'NORMAL' mode, the file 'main.c', and the function 'main'.

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 20
5 int days(int month, int year);
6
7 int main() {
8     int array[MAX];
9     int array1[MAX];
10    int array2[MAX];
11    int array3[MAX];
12    int sum = 0, sum2 = 0, sum3 = 0;
13
14    for (int i = 0; i < MAX; i++) {
15        array[i] = rand() % 150;
16        printf("%d ", array[i]);
17    }
18
19    for (int i = 0; i < MAX; i++) {
20        if (array[i] > 0 && array[i] < 50) {
21            array1[sum] = array[i];
22            sum++;
23        } else if (array[i] >= 50 && array[i] < 100) {
24            array2[sum2] = array[i];
25            sum2++;
26        } else if (array[i] >= 100) {
27            array3[sum3] = array[i];
28            sum3++;
29        }
30    }
31
32    printf("\nlista 1 (0-49):\n");
33    for (int i = 0; i < sum; i++) {
34        printf("%d ", array1[i]);
35    }
36    printf("\n");
37    printf("\nlista 2 (50-99):\n");
38    for (int i = 0; i < sum2; i++) {
39        printf("%d ", array2[i]);
40    }
41    printf("\n");
42    printf("\nlista 3 (100+):\n");
43    for (int i = 0; i < sum3; i++) {
44        printf("%d ", array3[i]);
45    }
46    printf("\n");
47    return 0;
48 }
```

3.4. Prueba de escritorio

A terminal window with a dark background and an anime-style illustration of a character's face on the right side. The terminal shows the execution of a program that prints three lists of numbers. The prompt is [emilio@user1012007][ej3]{main}\$.

```
[emilio@user1012007][ej3]{main}$ ./a.out
133 136 27 115 143 85 136 42 99 121 62 127 140 109 113 76 90 126 22
136
lista 1 (0-49):
27 42 22

lista 2 (50-99):
85 99 62 76 90

lista 3 (100+):
133 136 115 143 136 121 127 140 109 113 126 136
[emilio@user1012007][ej3]{main}$
```